[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

May-26-0482

CSPE-612(vi) (Soft Computing)

B.Tech. 6th (NEP)

Time : 3 Hours **Max. Marks : 60**

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from section A, B, C, D. Section E is compulsory.

SECTION - A

1. (a) Define Soft Computing and list its features. (4)
   (b) Explain fundamentals of Artificial Neural Networks. (4)
   (c) Explain characteristics of neural networks. (4)
2. (a) Define neuron model. Explain components of artificial neuron. (4)
   (b) Explain different activation functions used in ANN. (4)
   (c) Write short note on applications of ANN in real world. (4)

SECTION - B

3. (a) Explain Perceptron model. Discuss its limitations. (4)
   (b) Explain learning laws; Hebbian learning and Delta learning. (4)
   (c) Explain topology of neural network architecture. (4)
4. (a) Explain Multi Layer Perceptron (MLP) with neat diagram. (4)
   (b) Describe Backpropagation learning algorithm steps. (4)
   (c) Explain Auto-associative and Hetero-associative memories. (4)

SECTION - C

5. Explain ART network model and its types. Explain Counter Propagation network with working. (12)
6. Explain Support Vector Machine (SVM) classifier. Describe ensemble methods in supervised learning. (12)

SECTION - D

7. Explain fuzzy model and fuzzy rule base. Explain fuzzy inference methods (Mamdani and Sugeno). (12)
8. Explain Neuro-Fuzzy systems and their advantages. Write applications of Neuro-Fuzzy systems. (12)

SECTION - E (Compulsory)

9. (a) What is perception?
   (b) Define BAM.
   (c) What is reinforcement learning?
   (d) Define fuzzy relation.
   (e) What is membership function?
   (f) Define PCA. (6×2=12)